Bronchiolitis

Section 1: Case Summary

Scenario Title:	**Bronchiolitis and Respiratory Failure**
Keywords:	Pediatric, intubation, respiratory failure, RSV, bronchiolitis
Brief Description of Case:	Pediatric Respiratory Failure

Goals and Objectives	
Educational Goal:	To demonstrate the management of a critically ill patient with bronchiolitis.
Objectives: (Medical and CRM)	Medical: • Identify respiratory distress in an infant • Review the diagnosis of bronchiolitis, including evidence and non-evidence-based therapies, and distinguish it from asthma • Initiate supportive care with supplemental oxygen and fluids • Recognize treatment failure and intubate an infant CRM: • Use appropriate resources to check medication doses and equipment sizes • Demonstrate effective teamwork and communication skills in a pediatric resuscitation • Demonstrate graded assertiveness to identify a change in patient status
EPAs Assessed:	D1, F1, C1, C3

Learners, Setting and Personnel				
Target Learners:	☒ Junior Learners	☒ Senior Learners	☐ Staff	
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab	☐ In Situ	☐ Other:	
Recommended Number of Facilitators:	Instructors: 1			
	Sim Actors: 0			
	Sim Techs: 1			

Scenario Development	
Date of Development:	September 23, 2022
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Last Revision Date:	October 1, 2022
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Version Number:	5

Section 2A: Initial Patient Information

A. Patient Chart				
Patient Name: Zarah		Age: 6 months	Gender: F	Weight: 7.5 kg
Presenting complaint: Noisy breathing sent in by 811 (nursing hotline)				
Temp: 38.4C	HR: 180	BP: 86/52	RR: 60	SpO2 88% R/A
Cap glucose: 6.2 mmol/L		GCS: 15 (E4 V5 M6)		
Triage note: Recent viral illness. This AM, breathing sounded "noisy," so parents called 811 (telehealth nurse assessment service). Sent to the ED.				
Allergies: None				
Past Medical History: • Uncomplicated spontaneous vaginal delivery at term • No previous admissions to hospital • Up to date with vaccinations • No family history of asthma/atopy		Current Medications: None		

Section 2B: Extra Patient Information

A. Further History
Four days ago, an older sibling who recently started pre-school had a cold. The next day, 6-month-old Zarah fell sick. She has had a runny nose and cough but seemed to be doing fine until yesterday when she did not eat or drink very much. This morning, she had some noisy breathing, and her chest looked funny while she was breathing. When it did not go away after a couple of hours, Zarah's parents called 811 for advice. They were directed to go to the emergency department. Zarah has not urinated today. Her last bowel movement was yesterday evening, and it was normal. She is otherwise healthy and fully up to date with her vaccines.

B. Physical Exam	
List any pertinent positive and negative findings	
Cardio: Tachycardia. Capillary refill 4-5 seconds centrally, 6 seconds peripherally.	Neuro: Alert
Resp: Nasal flaring, grunting, suprasternal, intercostal, and subcostal indrawing are present. Crackles and wheezes on auscultation.	Head & Neck: Mouth appears dry
Abdo: Soft, unremarkable.	MSK/skin: Unremarkable

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin: Infant
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid

B. Special Equipment Required

- Intubation equipment
- Oxygen delivery equipment and tubing
- Equipment for a crystalloid bolus with the push-pull method (3 way stop-cock and syringe)
- MDI +/- spacer or nebulizer
- NIPPV and/or High flow nasal cannula

C. Required Medications

- Normal Saline
- Salbutamol for inhalation
- Epinephrine for inhalation
- Acetaminophen
- Ketamine
- Rocuronium

D. Moulage

None

E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor

F. Patient Reactions and Exam

Increased work of breathing
Crackles and wheezes on auscultation. (Crackles may need to be more prominent than wheezes to replicate auscultation of a bronchiolitic patient more accurately.)

If acetaminophen is given, decrease the temperature to 37.5 degrees.
If IV fluids are given, the heart rate will drop to 140.
If an epinephrine nebule is given, increase the heart rate by 10-15 bpm, do not change respiratory rate or oxygen saturation.
If salbutamol is administered, increase the heart rate by 10-15 bpm, do not change respiratory rate or oxygen saturation.
When the patient is started on oxygen, saturations increase to greater than 90%.

In stage two, after oxygen has been started, the patient will become apneic and exhibit central cyanosis. Oxygen saturation will decrease to 70%.
Once intubated, the cyanosis will resolve, and the oxygen saturation will improve to 95%.

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	*Description of role, expected behaviour, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)*
N/A	
N/A	

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus tachycardia HR: 180 BP: 86/52 RR: 60 O_2SAT: 88% T: 38.4 °C GCS: 15 Lungs: Diffuse crackles and wheezes	*The patient has severely increased work of breathing. The patient is coughing and has a runny nose.*	Expected Learner Actions ☐ Start oxygen ☐ Suction (copious amounts) ☐ Achieve IV access ☐ Give fluid bolus ☐ Check a bedside glucose ☐ Administer acetaminophen ☐ Consult respiratory therapy for hi-flow oxygen / NIPPV	Modifiers - Oxygen will improve saturation to greater than 90% - Fluid will decrease heart rate to 140 - Salbutamol or epinephrine will increase the heart rate by 10-15 points but will not change the work of breathing - Ipratropium will have no effect - Acetaminophen will reduce temperature to 37.5 degrees - Bedside glucose is 6.2 mmol/L Triggers - The patient has been started on oxygen, and all other expected actions have been satisfied. - The patient has been started on oxygen, and participants show no signs of giving fluid or acetaminophen.	Participants will likely try salbutamol, ipratropium, or epinephrine during this phase of the simulation. They will be ineffective. If participants attempt further interventions to treat asthma (magnesium), the facilitator may need to emphasize the lack of effect.

2. Apnea and Intubation Rhythm: Sinus tachycardia HR: 140 if fluids given BP: 86/52 RR: 0 O_2SAT: 70% T: 37.5 °C if fever treated	*Patient has an apneic episode requiring escalation to mechanical ventilation.*	Expected Learner Actions ☐ Identify apnea ☐ Call for help ☐ Select appropriate intubation equipment and medications ☐ Intubate patient	Modifiers - Apnea will cause oxygen saturation to drop to 70%, and central cyanosis will appear on the mannequin Triggers - Patient has been intubated	The difficulties of a pediatric airway are not the focus of this case.
3. Improvement following mechanical ventilation Rhythm: Sinus tachycardia HR: 140 if fluids given BP: 86/52 RR: 20 (or bagging rate/ventilator rate) O_2SAT: 94% T: 37.5 °C if fever treated	*The patient stabilizes following invasive ventilation*	Expected Learner Actions ☐ Initiate post-intubation sedation ☐ Call PICU at the pediatric referral centre	Modifiers - Intubation will improve oxygen saturation to 95%, and cyanosis will resolve Triggers - Case ends when PICU has been called	Ventilation settings are not the focus of this case.

Appendix A: Laboratory Results

CBC WBC 6.0 Hgb 130 g/L Plt 350 Lytes Na 135 mmol/L K 4.0 mmol/L Cl 99 mmol/L HCO_3 26 mEq/L AG 10 Urea 4.0 mmol/L Cr 30 umol/L Glucose 6.2 mmol/L Extended Lytes Ca 2.3 mmol/L Mg 0.7 mmol/L PO_4 2.0 mmol/L VBG pH 7.4 pCO_2 45 mmHg HCO_3 26 mEq/L Lactate 1.6 mmol/L	

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

https://www.bmj.com/content/375/bmj-2021-064132

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Compare and contrast respiratory failure with a more typical presentation of bronchiolitis.

Laboratory work and chest imaging are very reasonable investigations in this case. However, in straightforward bronchiolitis, they are unnecessary.

Review which treatments for bronchiolitis are evidenced-based.

Discuss strategies for distinguishing bronchiolitis from asthma.

Ensure participants have methods to select drug doses and equipment sizes safely.

Identify the Canadian Pediatric Society Guidelines as a key resource for bronchiolitis management. (https://cps.ca/en/documents/position/bronchiolitis)

References

1. Friedman J, Rieder M, Walton J. Bronchiolitis: Recommendations for diagnosis, monitoring and management of children one to 24 months of age [Internet]. Canadian Pediatric Society; 2021. Available from: https://cps.ca/en/documents/position/bronchiolitis